

IA: MACHINE LEARNING FOUNDATIONS

SESSION 4 – Regression

Adrián Sánchez Morales

AGENDA

1. Simple Linear Regression
   - 1.1. Estimating coefficients
   - 1.2. Accuracy of the coefficients
   - 1.3. Accuracy of the model

2. Multiple Linear Regression
   - 2.1. Regression coefficients
   - 2.2. Some important questions

3. Other considerations
   - 3.1. Qualitative predictors
   - 3.2. Extension of LR
   - 3.3. Potential problems



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THE LINEAR MODEL

- One of the simplest approaches for supervised learning (quantitative response).
- It is important having a good understanding of LR before studying more complex methods.

Advertising Dataset

1. *Is there a relationship between advertising budget and sales?*
2. *How strong is the relationship between advertising budget and sales?*
3. *Which media are associated with sales?*
4. *How large is the association between each medium and sales?*
5. *How accurately can we predict future sales?*
6. *Is the relationship linear?*

1. SIMPLE LINEAR REGRESSION

1.1. ESTIMATING COEFFICIENTS

- A very straightforward approach for predicting a quantitative response Y on the basis of a **single predictor variable** X.
- It assumes that there is approximately a linear relationship between X and Y:

$$Y \approx \beta_0 + \beta_1 X$$

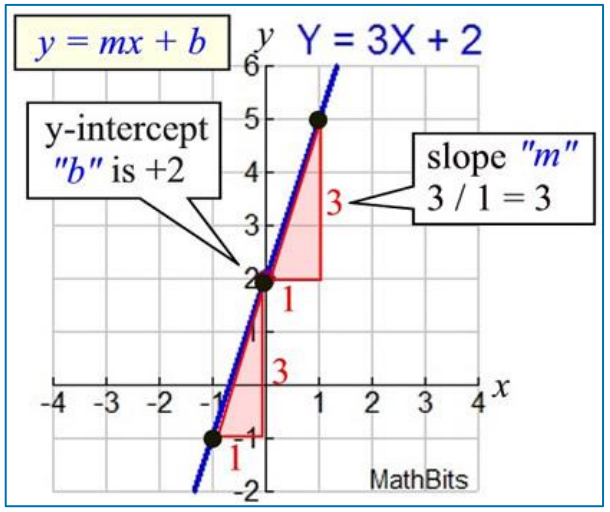
Parameters/Coefficients → *Intercept*, *slope*

- After training:

$$(x_1, y_1),\ (x_2, y_2), \ldots,\ (x_n, y_n) \qquad \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

- So, we want to find an *intercept* and a *slope* such that the resulting line is as close as possible to the *N* data points.

Linear Equation



1. SIMPLE LINEAR REGRESSION

1.1. ESTIMATING COEFFICIENTS

- Least Squares Approach:

Residual Sum of Squares $\text{RSS} = e_1^2 + e_2^2 + \cdots + e_n^2$

$$e_i = y_i - \hat{y}_i$$

$$\text{RSS} = (y_1 - \hat{\beta}_0 - \hat{\beta}_1 x_1)^2 + (y_2 - \hat{\beta}_0 - \hat{\beta}_1 x_2)^2 + \cdots + (y_n - \hat{\beta}_0 - \hat{\beta}_1 x_n)^2$$

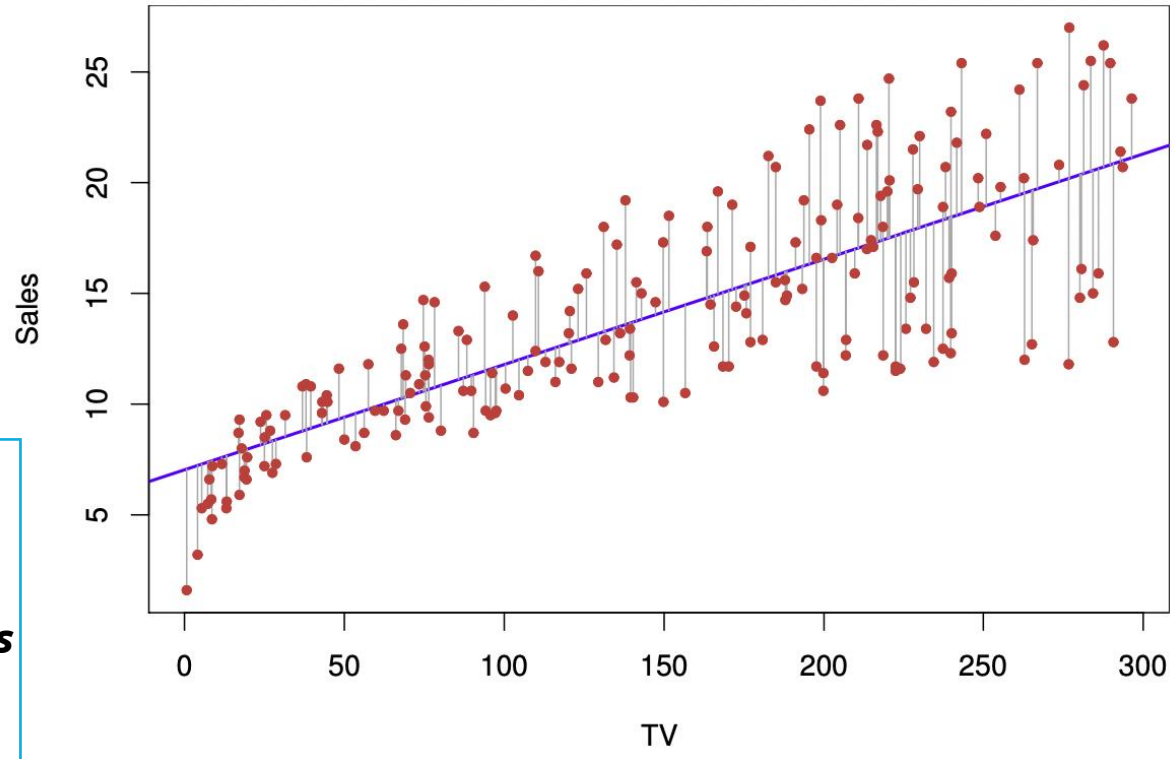
$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Least Squares Coefficients

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\bar{y} \equiv \frac{1}{n}\sum_{i=1}^{n} y_i \text{ and } \bar{x} \equiv \frac{1}{n}\sum_{i=1}^{n} x_i$$

Advertising Dataset



1. SIMPLE LINEAR REGRESSION

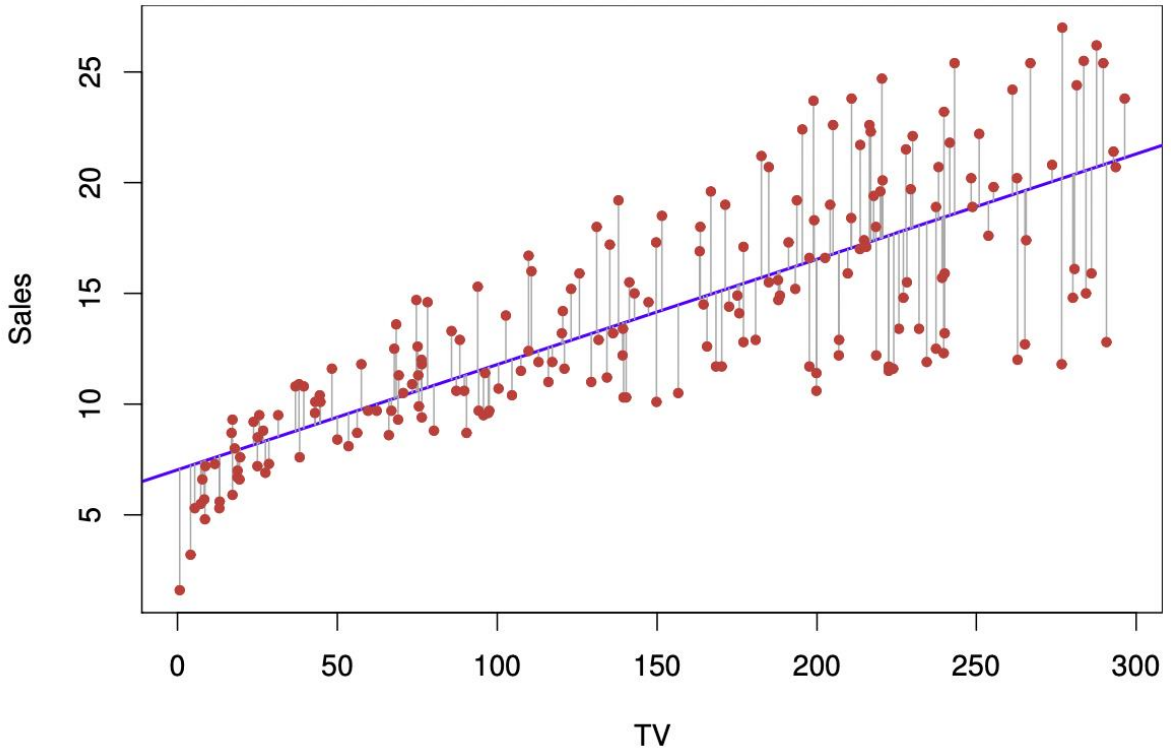
1.1. ESTIMATING COEFFICIENTS

EXERCISE

Write the python code that plots the figure to the right and estimates the *Least Squares Coefficients* for TV and Sales relationship of the Advertising Dataset.

How many product units will be sold if we spend 1000$ on TV advertising?

Advertising Dataset



1. SIMPLE LINEAR REGRESSION

1.1. ESTIMATING COEFFICIENTS

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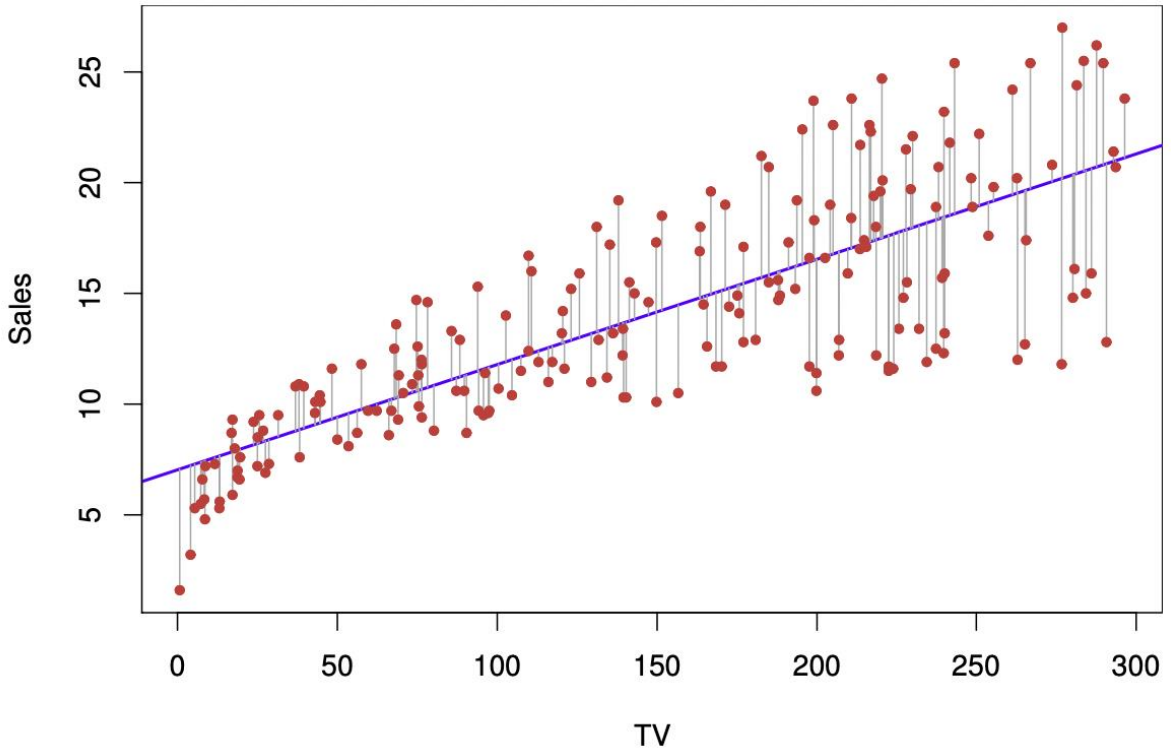
SOLUTION

$$\hat{\beta}_0 = 7.03$$

$$\hat{\beta}_1 = 0.0475$$

47.5 additional units of the product

Advertising Dataset



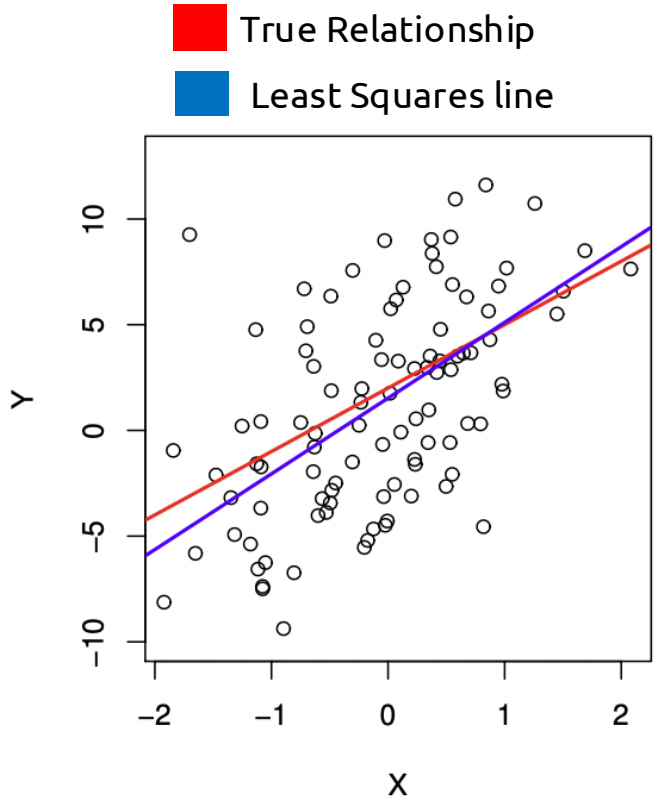
1. SIMPLE LINEAR REGRESSION

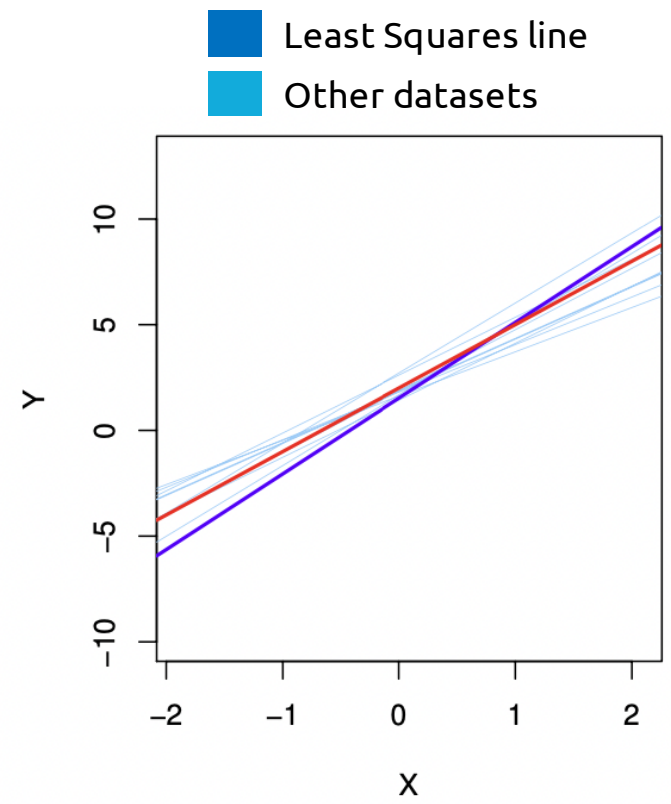
1.2. ACCURACY OF THE COEFFICIENTS

$$Y = f(X) + \epsilon \longrightarrow \boxed{Y = \beta_0 + \beta_1 X + \epsilon}$$

- We assume ϵ is independent of X
- In practice, the true relationship is not known, and the result will depend on the sample data used during training.
- Different data sets generated from the same true model result in slightly different least squares lines.







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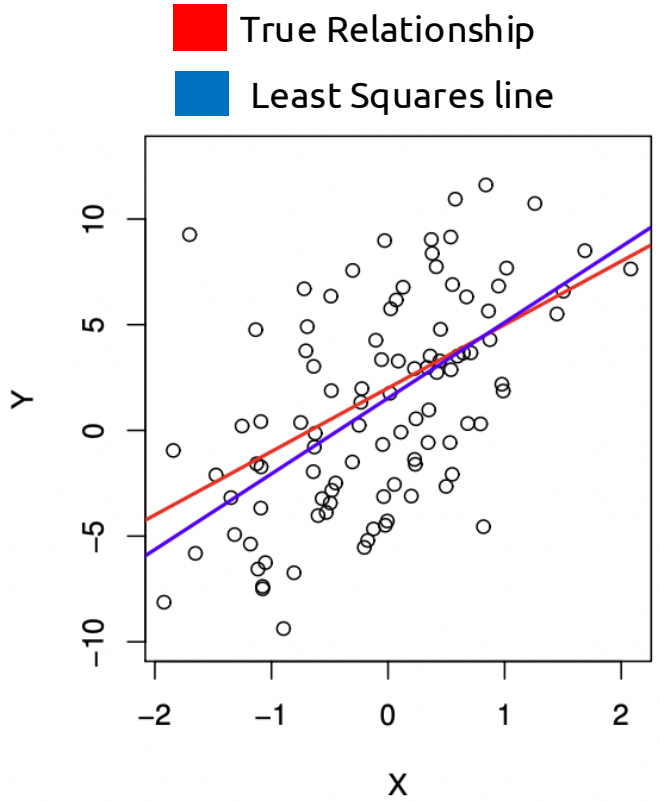
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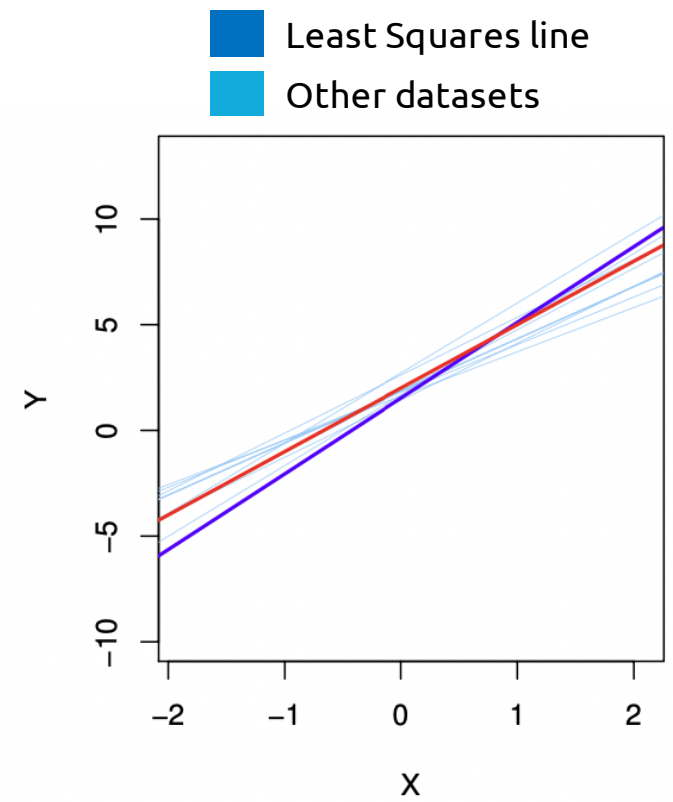
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Estimating characteristics of a large population from a sample.



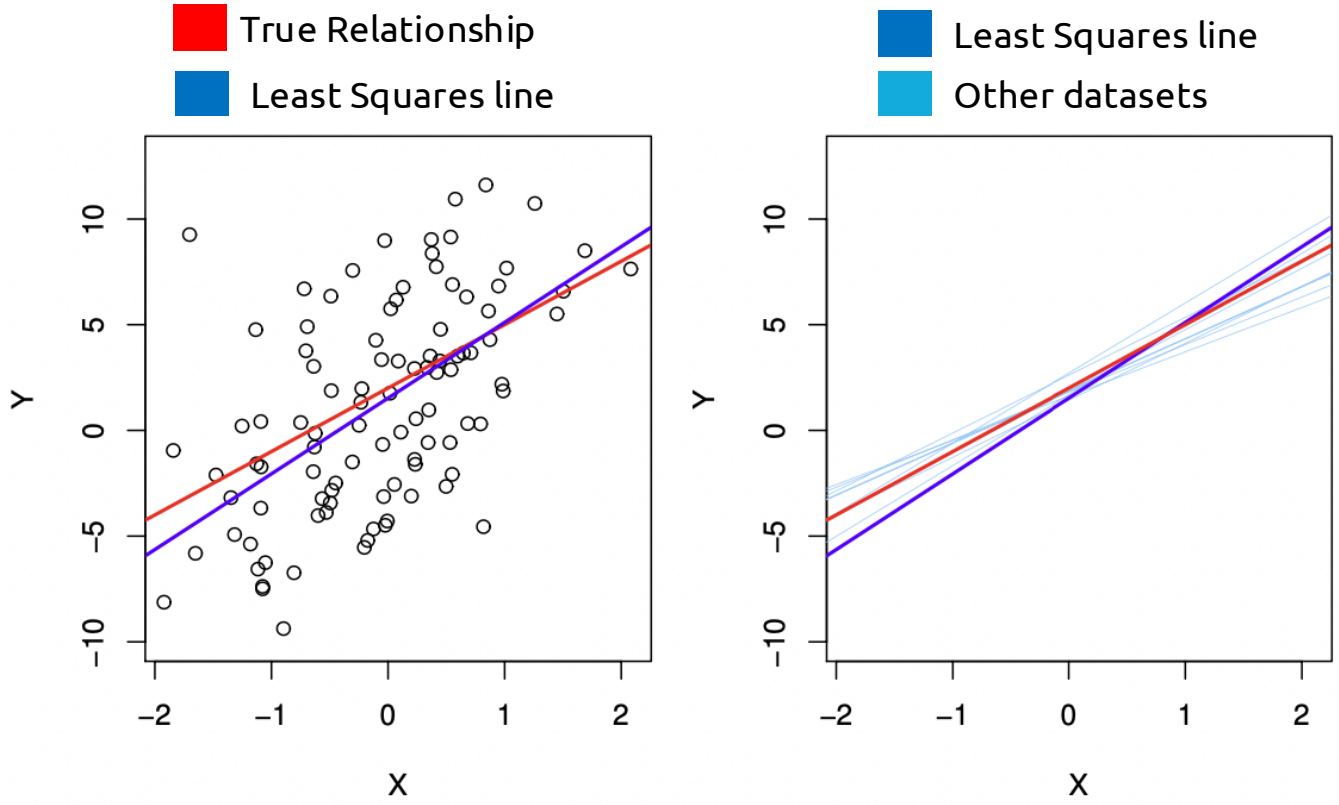


1. SIMPLE LINEAR REGRESSION

1.2. ACCURACY OF THE COEFFICIENTS

- Estimation of the mean of a random variable (Basic Statistics)
  - Estimating μ from a sample ⟶ Biased result:
    - Overestimate μ
    - Underestimate μ
  - If we could average the estimates obtained over a huge number of data sets, then the average of these estimates would be spot on!
- Having just a sample, how far are we?

Standard Error $\mathrm{Var}(\hat{\mu}) = \mathrm{SE}(\hat{\mu})^2 = \frac{\sigma^2}{n}$



1. SIMPLE LINEAR REGRESSION

1.2. ACCURACY OF THE COEFFICIENTS

$$\mathrm{SE}(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \right]$$

$$\mathrm{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

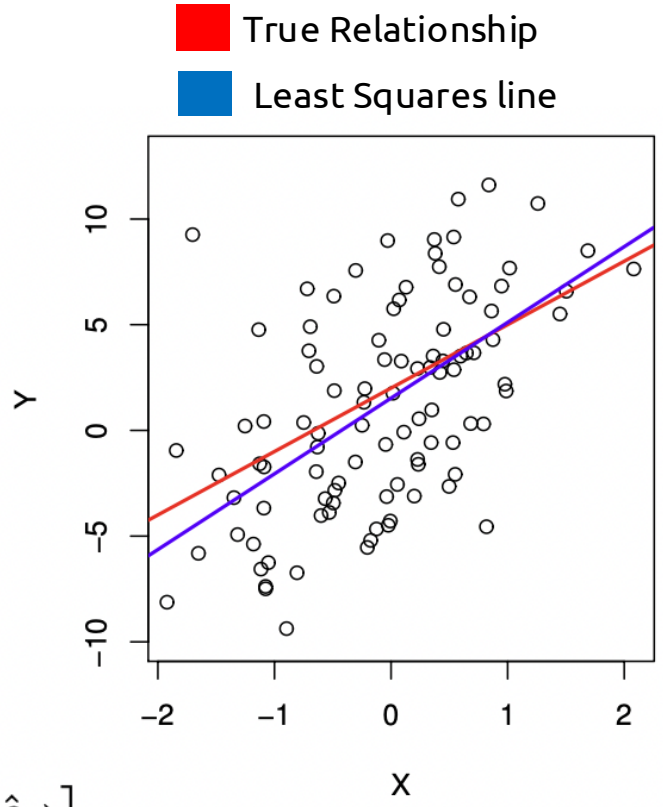
$$\sigma^2 = \mathrm{Var}(\epsilon)$$

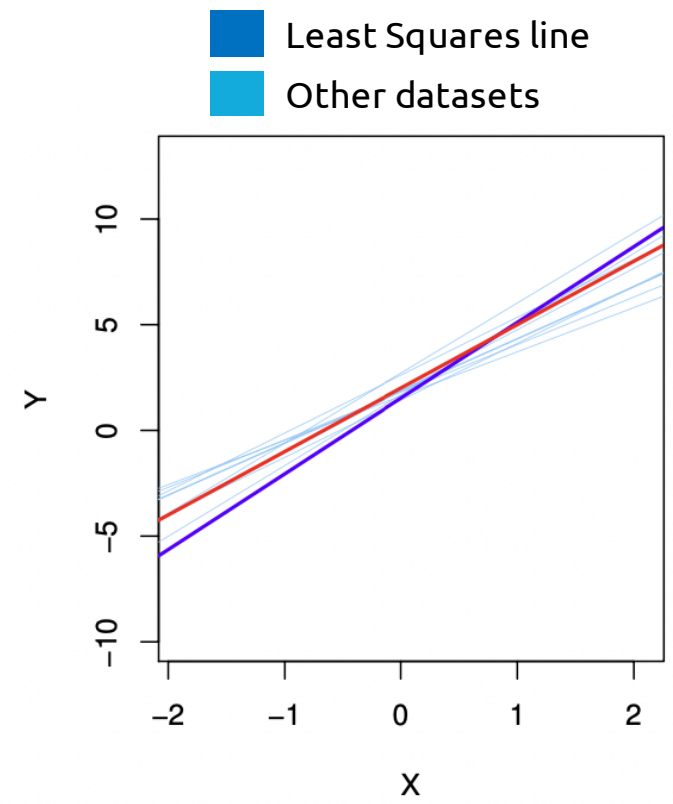
- Confidence Interval:

"A X% confidence interval is defined as a range of values such that with X% probability, the range will contain the true unknown value of the parameter"

- For linear regression, the 95% confidence interval for β_1 or β_0 :

$$\hat{\beta}_1 \pm 2 \cdot \mathrm{SE}(\hat{\beta}_1) \longrightarrow \left[\hat{\beta}_1 - 2 \cdot \mathrm{SE}(\hat{\beta}_1),\ \hat{\beta}_1 + 2 \cdot \mathrm{SE}(\hat{\beta}_1) \right]$$





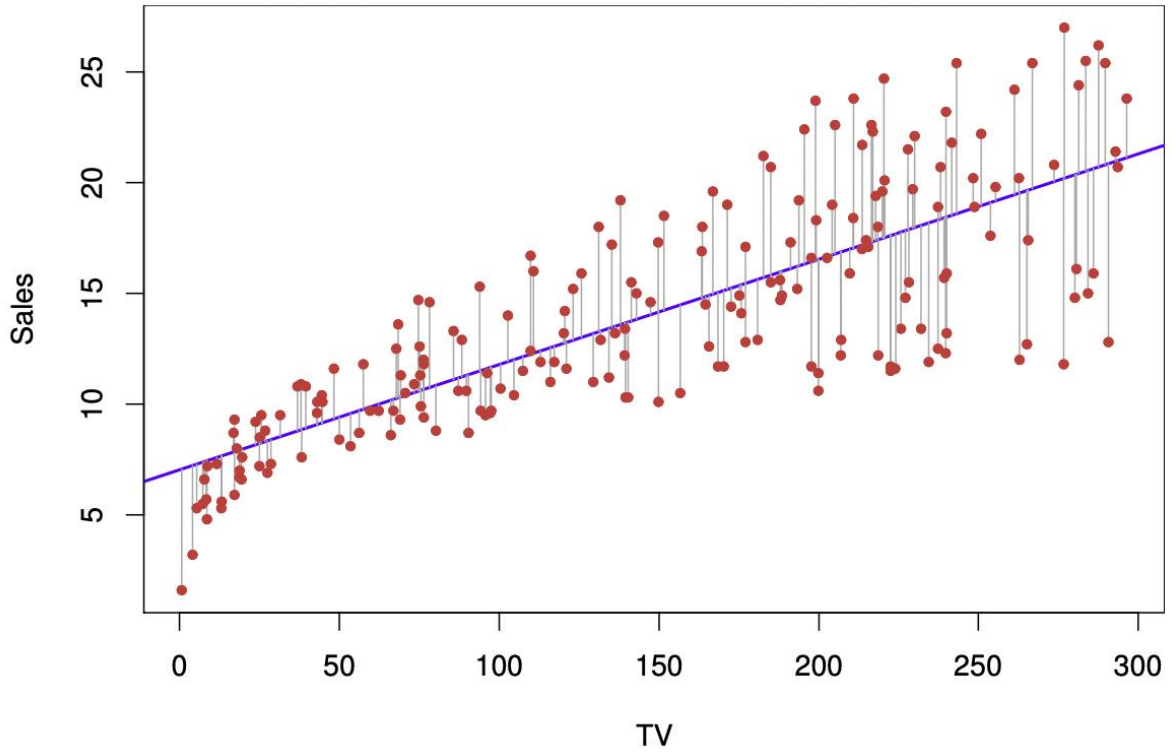
1. SIMPLE LINEAR REGRESSION

1.2. ACCURACY OF THE COEFFICIENTS

EXERCISE

Which are the 95% confidence interval for the parameters β_0 and β_1 previously calculated?

Advertising Dataset



1. SIMPLE LINEAR REGRESSION

1.2. ACCURACY OF THE COEFFICIENTS

EXERCISE

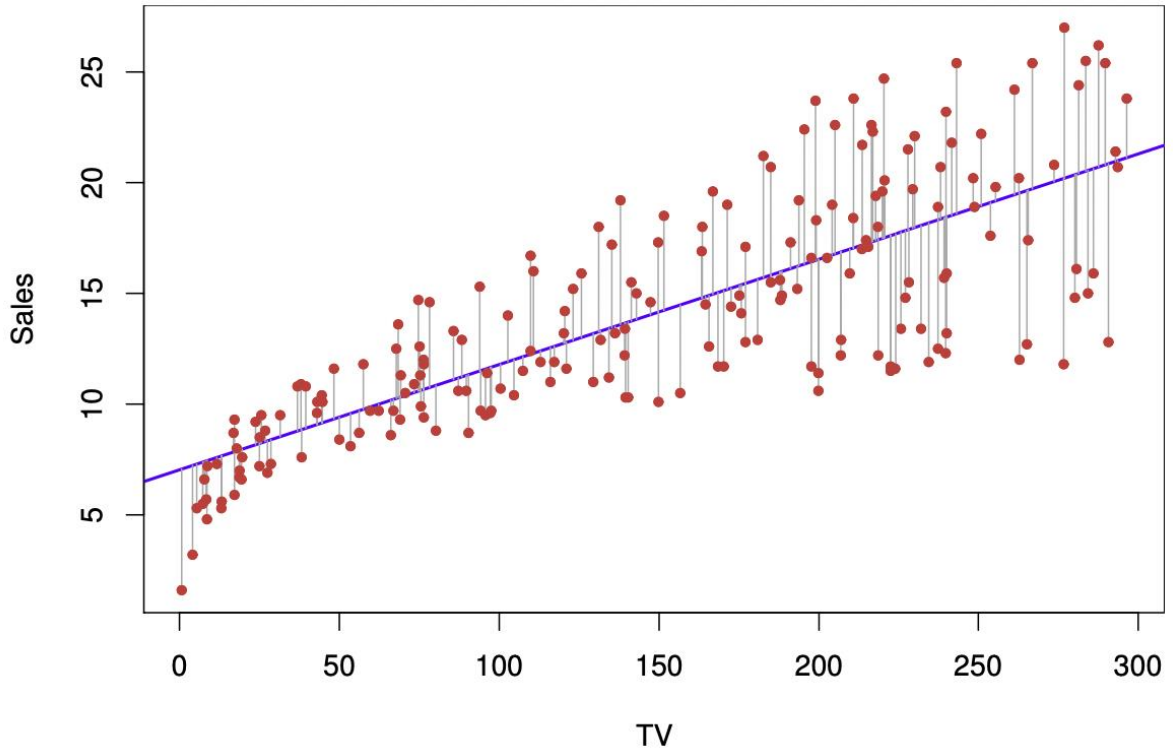
Which are the 95% confidence interval for the parameters β_0 and β_1 previously calculated?

SOLUTION

$$\beta_0 \longrightarrow [6.130, 7.935]$$

$$\beta_1 \longrightarrow [0.042, 0.053]$$

Advertising Dataset



1. SIMPLE LINEAR REGRESSION

1.3. ACCURACY OF THE MODEL

Residual Standard Error

The average amount that the response will deviate from the true regression line

$$\mathrm{RSE} = \sqrt{\frac{1}{n-2}\mathrm{RSS}} = \sqrt{\frac{1}{n-2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

$$\mathrm{RSS} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

R^2 Statistic

It is a proportion. It takes values between 0 and 1 and is independent of the scale of Y.

$$R^2 = \frac{\mathrm{TSS} - \mathrm{RSS}}{\mathrm{TSS}} = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}$$

Total Sum of Squares $\mathrm{TSS} = \sum(y_i - \bar{y})^2$

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3. Other considerations
   - 3.1. Qualitative predictors
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2. MULTIPLE LINEAR REGRESSION

2.1. REGRESSION COEFFICIENTS

- Regression model with more than one predictor variable.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon$$

- Why multiple regression if we can have separate simple linear regression models?

2. MULTIPLE LINEAR REGRESSION

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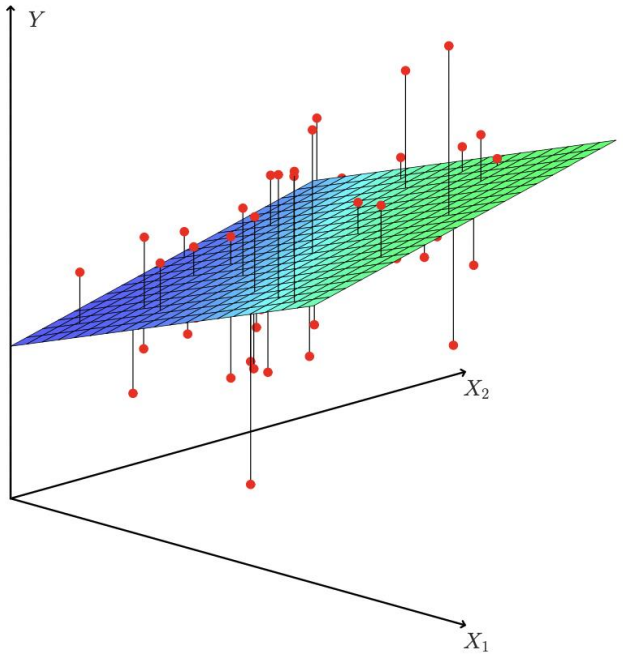
- Regression model with more than one predictor variable.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon$$

- Why multiple regression if we can have separate simple linear regression models?
  - Each model is associated with a different linear solution
  - Each individual regression ignores the others

- As was the case in the simple linear regression:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_p x_p \qquad \text{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

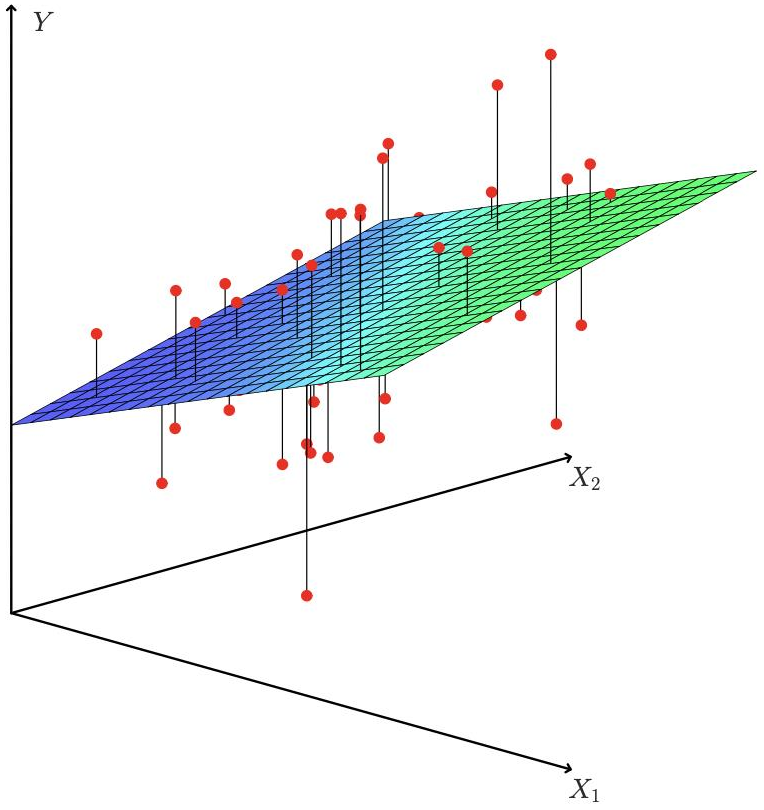


- Estimating the coefficients is similar to the SLR case.
- However, it is way more difficult to represent estimated coefficients.
- It's recommended to use a software package tool.

2. MULTIPLE LINEAR REGRESSION

EXERCISE

Write the python code that plots the figure to the right (or similar) and calculates the *Coefficients* for Sales estimation by using Radio and TV as predictors of the Advertising Dataset.



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3. OTHER CONSIDERATIONS

3.1. QUALITATIVE PREDICTORS

Only Two Levels

- Imagine that we wish to investigate differences in credit card balance between those who own a house and those who don't.

$$x_i = \begin{cases} 1 & \text{if } i\text{th person owns a house} \\ 0 & \text{if } i\text{th person does not own a house.} \end{cases}$$

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i = \begin{cases} \beta_0 + \beta_1 + \epsilon_i & \text{if } i\text{th person owns a house} \\ \beta_0 + \epsilon_i & \text{if } i\text{th person does not.} \end{cases}$$

β_0 as the average credit card balance among those who don't own

β_1 as the average credit card balance among those who own their house

OR

$$x_i = \begin{cases} 1 & \text{if } i\text{th person owns a house} \\ -1 & \text{if } i\text{th person does not own a house} \end{cases}$$

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i = \begin{cases} \beta_0 + \beta_1 + \epsilon_i & \text{if } i\text{th person owns a house} \\ \beta_0 - \beta_1 + \epsilon_i & \text{if } i\text{th person does not own a house.} \end{cases}$$

It is important to interpret the results correctly

3. OTHER CONSIDERATIONS

3.1. QUALITATIVE PREDICTORS

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More than Two Levels

- A "dummy" variable is created for each possible level

$$region = \{South, West\}$$

$$x_{i1} = \begin{cases} 1 & \text{if } i\text{th person is from the South} \\ 0 & \text{if } i\text{th person is not from the South} \end{cases}$$

$$x_{i2} = \begin{cases} 1 & \text{if } i\text{th person is from the West} \\ 0 & \text{if } i\text{th person is not from the West} \end{cases}$$

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i$$

- It is important to interpret the results correctly

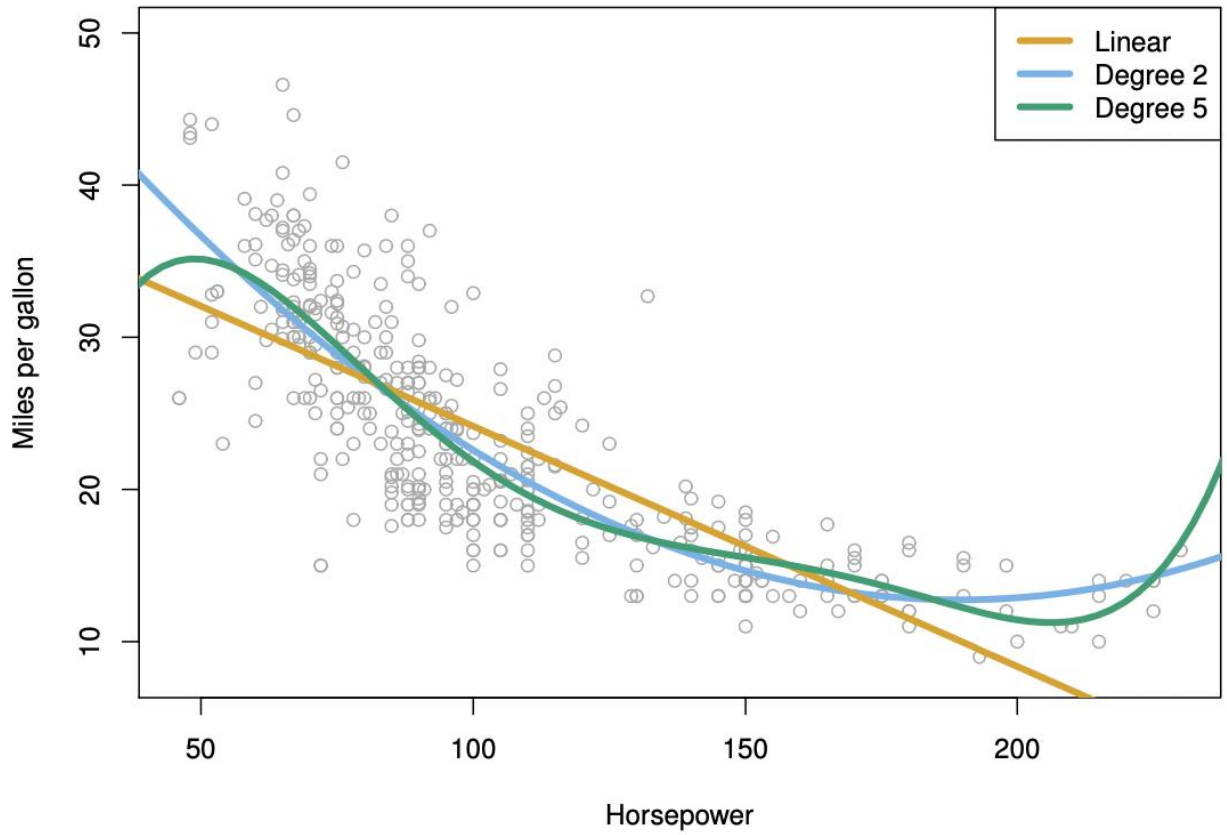
3. OTHER CONSIDERATIONS

3.2. EXTENSION OF LINEAR REGRESSION

- In some cases, the true relationship between the response and the predictors may be non-linear.

Polynomial Regression

$$\texttt{mpg} = \beta_0 + \beta_1 \times \texttt{horsepower} + \beta_2 \times \texttt{horsepower}^2 + \epsilon$$

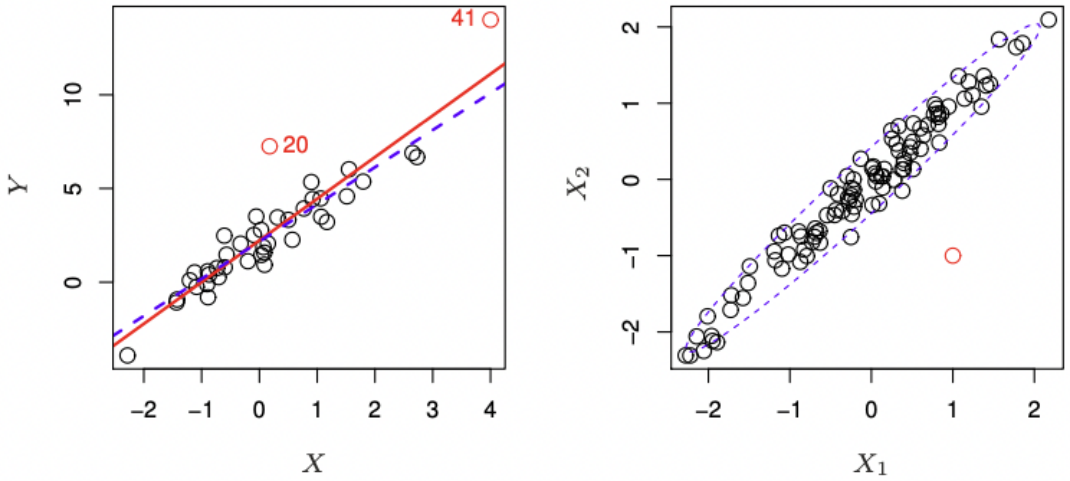


3. OTHER CONSIDERATIONS

3.3. POTENTIAL PROBLEMS

Outliers

- An outlier is a point for which y_i is far from the value predicted by the model
- Several reasons such as incorrect recording of observations during data collection
- Linear Regression models are sensitive to these outliers



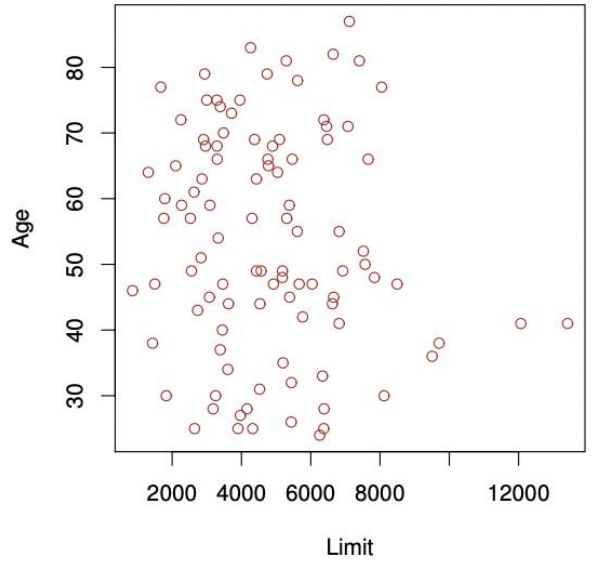
- There are several ways of detecting outliers:
  - Plots
  - Statistical tools
  - Algorithms

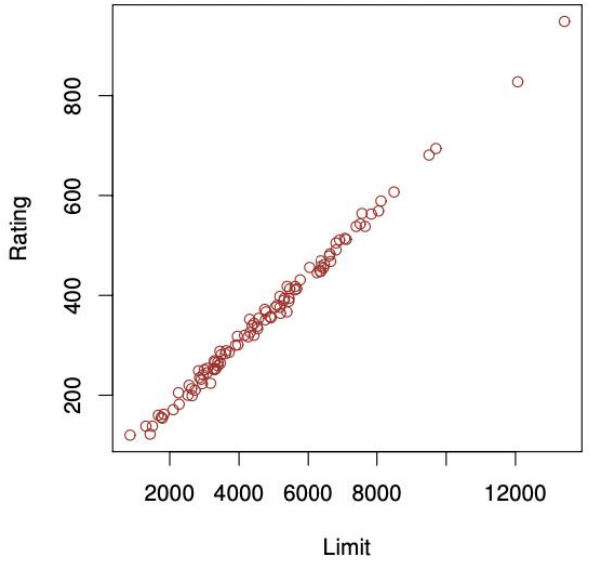
3. OTHER CONSIDERATIONS

3.3. POTENTIAL PROBLEMS

Collinearity

- Refers to the situation in which two or more predictor variables are closely related to one another.
- It may be difficult to separate the individual effects of collinear variables on the response





THE LINEAR MODEL

Advertising Dataset

1. *Is there a relationship between advertising budget and sales?*

Multiple LR model

2. *How strong is the relationship between advertising budget and sales?*

RSE and R^2

3. *Which media are associated with sales?*

Confidence Interval and p-values

4. *How large is the association between each medium and sales?*

Standard Error and Confidence Interval

5. *How accurately can we predict future sales?*

Confidence Interval

6. *Is the relationship linear?*

Residual Plots


THANK YOU